IBM-LASL meeting

STRETCH Meeting
May 9, 1957

adjenda

(1) Gen. Progress Report
(2) Review Exchange
(3) Programming & vocab.
(4) Ram-Exchange coop.
(5) I/O computer
(6) algebraic flow diags.

Los Alamos	IBM
Carlson	Dunwell
Frank	Sorahan
Woods	Johnson
Kolsky	Bucholtz
Walton	Herb Wild
Cherry	Jack Gibson
Blewett	
Goldstein	

general report (Dunwell)

Transistors
packaging } Engr. progress largely "submerged".
ventilation

Memory: data-flow models (bench models) ~ Sept 1 scheduled. close to conclusion now on ½ μs memo 3-hole device. main lack is driver Transistors (NPN) are now using tubes. driver transistors - have made exp. models. but

Transistors: (1) diffusion process is hard to control (2) measuring crystals to start with. "lucky crystals",

Time sequence: (1) PNP, (2) NPN Switching, (3) NPN Drivers
↑ working here now   feel confident

Circuits: orig sched. Transistor circs. ~ Dec '57. July '57 seems to be possible now) standards are set to take into account "ageing" & deviations of components. packaging, have checked sizable units at 10 mc.

Driving Transistors act at 32 as a body. ~ in 60 to 90 days can describe hole memory. may want to make boxes larger. ← (NOTE)

2 μsec Mem: Toroid cores. awaiting drivers

Exchange: near final freezing. should be running next year.

Bus system: making bench tests. not specific to particular machine. 2 methods are being considered now (Pomerene - asymetric opn.) ~60 days

adder: have 10 Mp adder, switching & registers are planned for July 1.

I/O Units: (take up later with Exchange)

Bu Ships: (1) support on Mem Units, (2) planning on their type of work.

Planning Groups specified 20 problems for future

IBM has given them a manual on machine. question of who owns it --

Basic Computer which can be changed to fit a variety of jobs.

- Exchange - memorys - I/O - etc. - so as not to do 3 indep. systems.

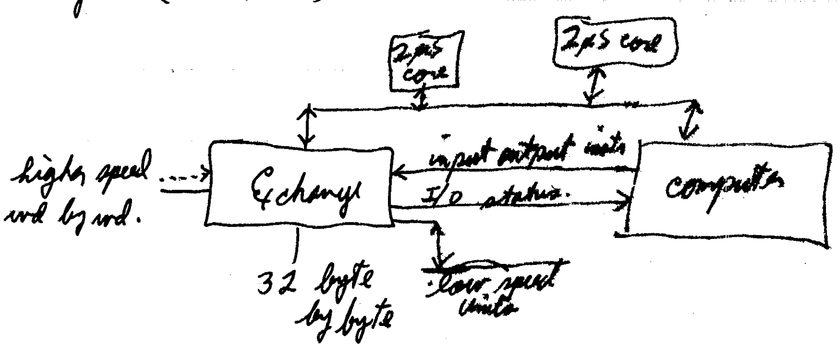
Decision to get machine. broad questions,

1. Cost
2. Delivery
3. how fast will do work, performance.
4. auto prog system.
5. reliability.

Exchange (Herb Wild)

Medium: 30 kc.

High speed: 30 words kc


2μS core
2μS core
higher speed word by word.
Exchange
input output units
I/O status.
computer
32 byte by byte
~~low speed~~ units

control wd

Data Wd address 20	Next control Wd addr. 20	Wd. count 15	Status 7	1	1

(index tag
not used error condition?

(Bucholtz)

Instructions for Exchange:

Read }
Write }
Locate
Control
Disconnect

Control Wd addr. 20	I/O unit addr 18	Index addr 12	opn etc.

Programmer should be able to think of each unit as separate — Exchange is passive
limit of 32 units (10 channels) if more are asked for — delay.

Select: computer waits for check whether unit is already selected before continuing.
REWind } a channel isn't assigned until unit says it is
Read } no longer busy.

Locate: to send addressing info. to I/O unit. e.g. Track or block on disk
coords for plotting.

Control: rewind, backspace, turn on lite, etc. I/O unit decodes field (1st 8 bits

Disconnect: "Safety valve" instr. cancels out order in Exchange. to free channels.
can program interrog. to see if we are where we want to be.

(1 Signals which are forced on

(3 Signals · programed interruption

Program Interrupt:

conditions. (1) Exchange cannot accept instr. Select Reject. Not ready reject. } come back immediately sets at transfer order, (unless masked out)

(2) any unit can give. (identify unit is carried along)

Normal End.
Operator Signal (defined by prog.)
End of File
Cancel
Data Error

} not correctable by auto cor. device.
e.g. keyboard wrong key struck

These will be stored away - can be interrogated any time. (They come back at odd times.)

Interrogate: 256 - 511 I/O addr.

can also be used as data wd.

Unit Status 7 bits : assign channel
data error
not ready
Select for read
Select for write
Select for locate
Select for control

77 bits total here.

1 bit Interrupt bit (used by EXChange)

Scanning of Exchange:

(1) channel scanner. (Top priority)
(2) instruction scan
(3) memory scan

grouping + distribution ?

questions:

1. stepping by other than 1, (±Δ)
2. read backwards?
3. automatic retry? —better to

(Buchholz) I/O Units

1. Card Reader ~1000 cards p min

   2 reading stations read & compare before sending to computer.

   10 bytes of 8 each 12 times for card.

   ∴ card image: 15 words 960 bits

   I/O computer can rearrange card image?

2. Card Punch ~155 cards/min

   1 punching station, 1 reading station (good for new cards only)

   960 bits, card image.

3. Mechanical Printer: 500 line/min wire printer at present (120c/line) hope to make it 1000 or more later.

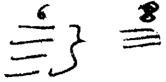
   Same as 720 wire printer.

4. Tape Units: 727 type (15,000. 6 bit characters/sec)

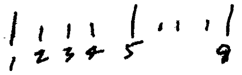
   729

   are handled as 8 bit bytes xxxxxx00

   may later take (compacting)



prints



1 5 9

then 2 6 10

3 7 11

4 8 12

4

3

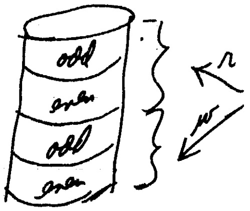
2

1

5. Inquiry station

   keyboard, typewriter, console sws. + lites (Power on, Power off, Program load. not part of console)

Disk: (Burnwell)



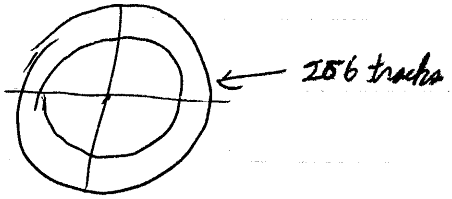
4 active sections, 1 set of arms for each. (either read or write)
2 sets
about 18 arms each
4 words
with 18 get 72 bits in 4 successive bit times.

capacity 4 million words
1 word / [illegible]8 µs

read and write

Note

1 wd read and write 1 wd in 8 µs
eg read in section 1, write in sect 2
(or vice versa)



4,096 wds. per track on 18 tracks.

go on [illegible] at beginning of a block only.

present Thinking: 1024 wds is 1 block length.

18,00[illegible] rpm
30. rpsec.

32 msec per rev.

Time between tracks less than one rev.

Can use

Pough.
June 12, 13 Wed, Thurs

→ 1. division of comp. sections labor between
2. problems for examination in detail

Fri, M

Discussion of large 3D probs.

Timing (300 orders/pt)$(1\times10^{-6}$ sec/order$)(2\times10^{5}$ pts$)$ = 60. sec

disk $(2.0\times10^{6}$ orders$)$ (8 μs/ord) = 16 sec

$$\phi' = \phi_{old} + \omega\left(\sum c_i \phi_j + \delta - \phi_{old}\right)$$

Reutos Eq.

~18 μs

7 quantities

ϕ_{old} c_i, S

Eg. Implicit eq:

$$\dot{R}^{n+1} = \ddot{R}\Delta t = \dot{R}^{n}$$

Implicit.

$$\dot{R}^{n+1} = \frac{\ddot{R}^{n} + \ddot{R}^{n+1}}{2}\Delta t + \dot{R}^{n}$$

$$R^{n+1} = R^{n} + \frac{\dot{R}^{n} + \dot{R}^{n+1}}{2}\Delta t - \frac{\ddot{R}^{n} - \ddot{R}^{n+1}}{12}\Delta t^{2}$$

1st [illegible]

Nearest Neighbor Calc. timing.

1. Basic eqs. (conservation laws) two cycles { mass - momentum I C_I / energy - thermo. II C_{II}
2. neighbor
   - I. – neighbor tape (call in one by one) ~~Time~~: call time / ~~Storage~~: (no. neighbors) } negligible
   - – call in block containing PS for m points → ~~Storage~~ n. ~~time~~: (1 disk rev. –?)
   - – go thru External mem call in neighbor quantities for m points

     Time: (one Disk rev.) $\frac{N}{m}$ per cycle.

     Storage: (10 m)(one no. neighbors/pt.)
   - – calc mass + momentum eqs. for m pts

     time: $m C_I$

     storage: code$_I$ [illegible]

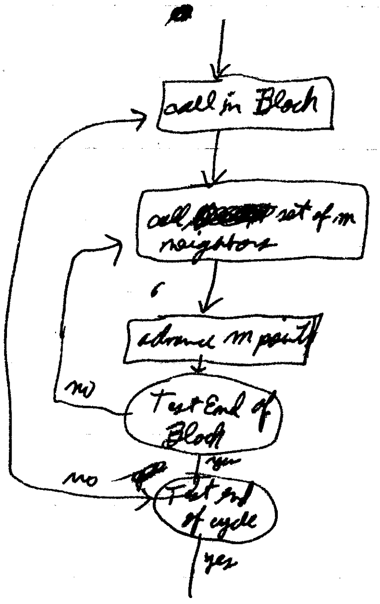
n = no. in block

m = no. advanced at a time

$m < n$

but an integral multiple

Tests to repeat



① Total Time:

$$\frac{N}{n}(1 \text{ disk time} -) \quad \text{read in of P.S.}$$

$$+ \frac{N}{m}(1 \text{ disk time})\left(\frac{N}{m}-1\right) \quad \text{call in neighbors}$$

$$+ N C_I$$

[(Eg.) $N = 10^4$, disk Time = $3. \times 10^{-2}$ sec
$n = 10^3$, $m = 10^2$, $C_I = 3 \times 10^{-4}$ sec]

$$10(3 \times 10^{-2}) + 10^2(3 \times 10^{-2})(10 - 1) + 3.0$$

$$0.3 + \overset{\text{[illegible]}}{270} + 3 = \overset{30.3}{\text{[illegible]}} \text{ sec}$$

note:

II. – call in block storage: n, time (1 disk)

– calc energy thermo. storage: code$_{II}$ time $n C_{II}$

② – repeat for each block

total Time: $\frac{N}{n}(1 \text{ disk time})$

$+ C_{II} N$

$C_{II} = 3 \times 10^{-4}$

$$10(3 \times 10^{-2}) + 3.0 = 3.3 \text{ sec}$$

Total for cycle: 9.6 sec { I/O = 30.3 [illegible] sec / calc = 6.0 sec

Neighbor Search:

~~- make pass~~ [illegible] ~~manufacture i, ξ, η~~ ~~Time: 1 disk rev.~~

I, Form graph ~~Storage: N~~

~~- Sort on~~ [illegible] ~~$\xi\eta$ comparing to previous $\xi\eta$~~

N_s = no of graph wds. — call in graph section : l wds (1 rev. time) Time: storage ③

l = no per section: [illegible] — call in wds form $\xi\eta$ & PW_i Time: (1 rev time)

II. list neighbors.

call in graph sections. Time: 1 rev.

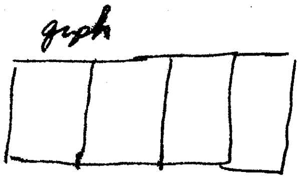
write part of neighbor ~~[illegible]~~ block Storage: l

calculate range Time: 1 rev. ④

C_{III}

Total Time : $\frac{N_s}{l}(1\,\text{rev}) + \frac{N_s}{l}(1\,\text{rev})\frac{N}{n} + \frac{N_s}{l}(2\,\text{rev}) + N C_{III}$

graph



Eg: $N_s = 4\times10^4$, $l = 10^4$, $C_{III} = 10^{-4}$

$(4)(3\times10^{-2}) + (4)(10)(3\times10^{-2}) + (4)(6\times10^{-2}) + 1$

$.12 + 1.2 + .24 + 1 = 2.56$ sec

point cloud = 9.60

done only occasionally? → neighbor search = 2.56

12.16 sec

discussion

interference with I/O computer + Exchange, bus system

perhaps I/O mainly logic less lookahead 2µs/opn

H.S. [illegible] arithmetic

Must decide on organization ...

- question of a scale of "cost" transistors, registers,
- complexity of I/O + H.S.

evaluate in terms of problems – auto prog.

Basic Communication: Multiple computers can be hooked together ← basic principle